

Hongpeng Chen

Ph.D. Researcher | Robot Learning • Robotic Manipulation • Embodied AI

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Ph.D. researcher developing **reliable robotic manipulation** through learning from demonstration, multimodal planning, and safe skill adaptation, with hands-on perception–planning–control integration on physical robot manipulation.

EDUCATION & RESEARCH AFFILIATIONS

The Hong Kong Polytechnic University, Hong Kong SAR Ph.D. in Industrial and Systems Engineering; advisor: Prof. Pai Zheng	Sep 2023 – Jul 2027 (expected)
Idiap Research Institute / EPFL, Switzerland Visiting Ph.D. Researcher in Robot Learning; advisor: Prof. Sylvain Calinon	2026
The Hong Kong Polytechnic University, Hong Kong SAR M.Sc. in Mechanical Engineering; advisor: Prof. David Navarro-Alarcon	Sep 2021 – Jun 2023
Shanghai Maritime University, Shanghai, China B.Eng. in Mechanical and Electronic Engineering; rank 3/72 (top 5%)	Sep 2016 – Jun 2020

TECHNICAL SKILLS

Programming / ML: Python, C++, MATLAB, PyTorch, JAX.

Learning / planning: Imitation learning, learning from demonstration, TAMP, VLM/LLM agents, behavior trees, grasp/motion planning.

Perception: RGB-D, 6D object pose estimation, Grounding DINO, AnyGrasp, OpenCV, PCL.

Robotics / control: ROS, force/impedance control; UR5e, Franka Emika Panda, RealSense D435i, 6-axis force/torque sensors.

Systems / simulation: Linux/Ubuntu, NVIDIA Isaac Sim, MuJoCo, Box2D.

SELECTED RESEARCH EXPERIENCE

GLaMP: Grounded Multimodal Agents for Robotic Manipulation 2024 – 2026

- Developed a grounded multi-agent framework combining pretrained VLM/LLM reasoning with a **PDDL-defined skill library** to translate natural-language instructions into executable behavior trees without task-specific training.
- Built the perception–execution stack with Grounding DINO, RGB-D geometric reasoning, 6D pose estimation, AnyGrasp-based grasp generation, and VLM verification for **closed-loop robot manipulation**.
- Evaluated **five real-world manipulation scenarios**, including pose perturbations and execution failures, with **78–89.5% task success** across tasks; published in *IEEE TIE*, 2026.

One-Shot Learning from Egocentric Demonstrations 2026 – Present

- Developed a one-shot framework that converts a **single egocentric human demonstration** into object-centric task constraints and executable long-horizon manipulation plans.
- Combined VLM grounding, structured task representations, and **task-and-motion planning (TAMP)** to handle scene variations without collecting new task-specific robot demonstrations.
- Implemented recovery for changed initial poses, displaced objects, and new obstacles; evaluated stacking, drawer manipulation, and multi-object placement on a **UR5e with RGB-D sensing**.

Safe Policy Adaptation for Imitation-Learned Robot Skills 2026 – Present

- Developed a dual geometric-field framework combining an imitation-learned nominal skill with a configuration-dependent safety field for obstacle avoidance while preserving task-relevant motion.
- Formulated safety modulation with a **task-dependent metric** to reduce deformation of the demonstrated skill relative to Euclidean modulation.
- Completed **2D benchmarks in JAX/Box2D** and **7-DoF manipulator simulations**, evaluating obstacle avoidance and motion preservation in higher-dimensional robotic configuration spaces.

RESEARCH EXPERIENCE (CONTINUED)

Force-Aware Learning for Contact-Rich Manipulation

2026 – Present

- Developing a hybrid pipeline using learned free-space motion for approach and force- and uncertainty-triggered local adaptation during contact, targeting insertion and surface-contact tasks.
- Integrated **6-axis force/torque sensing** with real-time robot control for contact-transition detection and corrective behaviors during contact-rich execution.

Demonstration-Based Impedance Control for Robotic Ultrasound

Oct 2022 – Jun 2023

- Learned variable impedance gains from human demonstrations using **GMM/GMR**; integrated them with a **constrained quadratic-programming controller** for real-time, ultrasound-guided scoliosis assessment.
- Evaluated the system in simulation, on spine phantoms, and with human participants to assess adaptive physical human–robot interaction.

Multi-Fidelity Bayesian Optimization for Robotic Assembly

May 2024 – Jan 2025

- Developed **multi-fidelity Bayesian optimization** with an LLM as a low-fidelity surrogate and a latent-variable Gaussian process linking its predictions to high-fidelity simulations for human–robot assembly.
- Implemented residual-based uncertainty quantification and variance-controlled LLM querying; evaluated robotic drilling for aircraft-cabin assembly; published at *AIM*, 2025.

LLM-Empowered Low-Code Programming for Industrial Robots

2024 – 2025

- Implemented a multi-agent natural-language programming workflow combining intent recognition, parameter parsing, and human verification to interpret operator instructions for robotic tasks.
- Integrated **retrieval-augmented assistance** from manuals, historical logs, and environmental data; demonstrated aircraft-panel drilling with non-expert operators; published in *Journal of Manufacturing Systems*, 2025.

Robotic Surface Inspection with Coverage Path Planning

2021 – 2025

- Implemented RANSAC- and curvature-aware region segmentation and local scanning-path generation for robotic line-scan inspection of free-form 3C components.
- Formulated global inspection coverage as a **constrained traveling-salesman problem** solved by particle swarm optimization; validated in simulation and on physical components; published in *IEEE TIM*, 2025.

SELECTED PUBLICATIONS

- [1] **H. Chen** et al. “GLaMP: A Grounded Language Model-Based Multiagent System for Long-Horizon Robotic Task Planning in Industrial Settings.” *IEEE TIE*, 2026.
- [2] **H. Chen** et al. “Multimodal Autonomous Robotic Long-Horizon Task Planning via Embodied Language Model and Behavior Trees.” *IROS*, 2025.
- [3] **H. Chen** et al. “Human-in-the-Loop Robot Learning for Smart Manufacturing: A Human-Centric Perspective.” *IEEE T-ASE*, 2025.
- [4] **H. Chen** et al. “PSO-Based Optimal Coverage Path Planning for Surface Defect Inspection of 3C Components With a Robotic Line Scanner.” *IEEE TIM*, 2025.
- [5] Z. Li, **H. Chen**, et al. “Deriving Symbolic Task Structure from a Single Egocentric Demonstration for Long-Horizon Task and Motion Planning.” *IEEE RA-L*, **under review**.
- [6] S. Liu, **H. Chen**, et al. “Scene-Grounded Persistent Prefix Consensus for Adaptive Execution Commitment in Vision-Language-Action Models.” *AAAI 2027*, **under review**.
- [7] **H. Chen**, et al. “Phase-Quotient Geometry for Minimum-Deformation Safe Imitation in Robot Manipulation.” *IEEE Transactions on Robotics*, **In Preparation**.

SELECTED HONORS

- 2025 IROS-SDC Travel Award
- PolyU Postgraduate Research Scholarship (2023-2027)
- PolyU Departmental MSc Dissertation Scholarship in 2022 (HK\$60,000 in total)
- Outstanding Undergraduate Thesis Award in 2020